MILITARY SPECIFICATION

INCREMENT, PROPELLANT, XM90

1. SCOPE

1.1 This specification covers one type of propellant increment designated as XM90 for Mortar Cartridge 81MM.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 —Strapping, Flat; Steel

MILITARY

MIL-B-2427—Boxes, Ammunition Packing; Wood, Nailed

MIL-A-2550—Ammunition & Special Weapons, General Specification for

MIL-B-3106—Board, Compositions, Water-Resistant, Solid.

MIL-L-10547 —Liners, Case, Waterproof

MIL-B-13239 —Barrier Material, Waterproofed, Flexible all Temperature.

MIL-I-45208—Inspection Requirements, General Specification for

MIL-I-45607—Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 —Inspection Terms and Definitions

DRAWINGS

ORDNANCE CORPS

20-4-613 —Marking Diagram and Sealing for Wooden Packing Boxes.

76-4-59 —Box, Packing, Metal-Wood, M18 for Smokeless Powder Assembly.

20-4-619 —Marking Diagram and Sealing of Fiberboard Cartons for Shipment of Propellant Increment.

76-4-46 —Box, Packing, with Metal Liner M24 for Smokeless Powder. Assembly and Details.

76-4-56 —Box, Packing, Metal-Wood, M17 for Smokeless Powder

7548645 —Carton, Packing, Reusable Collapsible for High Explosives, Assembly Details, Packing and Marking.

FSC: 1375

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7549033 —Container, Metal, Universal, M25 for Propellants and Explosives.

8858577 —Marking Diagram and Sealing of Container, Metal, Universal M25 for Shipment of Propellants.

8858848 —Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellants.

XP–113501 —Charge B, Propellant Increment, XM90: Mortar Cartridge 81MM.

XP–113502 —Bag, Propellant.

XP–113503 —Charge A, Propellant Increment XM90: Mortar Cartridge 81MM.

XP–113504 —Bag, Propellant.

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11 —Procedures and Tables for Continuous Sampling by Attributes.

IEL–8872199—Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71–90 —Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–90 (latest revisions)".

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Increments. The increments shall comply with all requirements specified on Drawing (dwg.) XP–13501, XP–13502, XP–13503 and XP–13504 and with all requirements specified in applicable specifications.

3.3 Workmanship. All increments shall be free of dirt, grease and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information

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accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of propellant increments produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain propellant from not more than one lot.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 *Loading Assembly (see dwg. 113501, 113502, 113503 and 113504).*

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.3)*
Critical:			
1.	Propellant exposed or escaping from Bag	Visual	01001
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Length between button holes	Gage	01002
202.	Length of button hole minimum	Gage	01003
203.	Location of button hole	Gage	01004
204.	Thread end length of cross seam incorrect	Gage	01005
205.	Stitches broken or missing	Visual	01006
206.	Evidence of poor workmanship (see 3.3)	Visual	01007

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4.2.2.2 *Sealed inner liner (see 5.1.1, 5.1.2, or 5.1.3).*

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Liner damaged so that contents are exposed or liable to become exposed	Visual	02001
102.	Bag incompletely sealed	Visual	02002
Minor: None defined			

4.2.2.3 *Sealed wood packing box (see 5.1.1, 5.1.2, or 5.1.3).*

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Strapping missing, broken or loose	Visual-Manual	03001
102.	Board broken or split	Visual	03002
103.	Contents exposed	Visual	03003
Minor:	AQL 1.50 percent		
201.	Strapping improperly engaged	Visual-Manual	03004
202.	Contents move when shaken	Manual	03005
203.	Marking misleading or unidentifiable	Visual	03006

4.2.3 *Testing.*

4.2.3.1 *Weight of propellant; under minimum Critical defect; over maximum Major defect (see dwg. 113501 and 113503).* The propellant increment weight shall be determined 100 percent and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weight determination, and if performed manually shall be performed by a different operator. Any propellant increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4. Code No. 04001.

4.2.3.2 *Air pressure test (Major defect).* The containers shall be tested 100 percent in accordance with 4.3.1. If any container fails to meet the requirement it shall be rejected and removed from the lot (see dwg. 8858848). Code No. 05001.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Lists Number IEL–8872199 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.2.

4.2.4.1 Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3. Test methods and procedures.

4.3.1 *Air pressure test for containers (see dwg. 8858848).* The container shall be subjected to an internal pressure of ½ to 1 pound per square inch by a method satisfactory to the contracting officer's representative. A water manometer shall be assembled in the system. A drop of 0.7 inches or more, on the manometer in 15 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.1)

5.1.1 *Level A.* Propellant increments shall be packed in wooden packing boxes that conform to Grade A, Type I, Class 1, Specification MIL–B–2427, except that bottoms, sides and top may be ½ inch thick lumber. The inside dimensions of the box shall be 14½ plus ⅛ inch long by 14½ plus ⅛ inch wide by 7¼ plus ⅛ inch deep. Each wooden packing box shall be fitted with a caseliner conforming to Type 1, Class 2, Grade A, Specification MIL–L–10547, manufactured from barrier material conforming to Grade A, Type CW–2, Specification MIL–B–13239. The caseliner shall be approximately 14⅞ inches long by 14¾ inches wide by 18½ inches deep. Both ends and top of wooden box shall be lined, between the box and the caseliner, with fillers manufactured of fiberboard conforming to Type I or II, Specification MIL–B–3106. The cover shall be secured to the box with nails or wood screws in accordance with Specification MIL–B–2427. The box shall be strapped girthwise, between

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cover cleats with two 5/8 inch by .02 inch thick flat steel straps conforming to Type 1, Class A or B, Specification QQ–S–781. The net weight shall not exceed 25 pounds of propellant.

5.1.2 *Level B.* Propellant increments shall be packed in propellant packing boxes M17, M18, M24 or M25 complying with dwg. 76–4–46, 76–4–59, 76–4–56 or 7549033 respectively. Containers shall be closed, sealed and tested in accordance with dwg. 8858848.

5.1.3 *Level C.* Propellant increments shall be packed in accordance with dwg. 7548645, Carton Number 1, except that the bag liner shall be manufactured from barrier material conforming to Grade A, Type CW–2, Specification MIL–B–13239.

5.2 Marking.

5.2.1 *Level A.* Exterior containers shall be marked in accordance with dwg. 20–4–613. The ICC nomenclature to be applied shall be "Propellant Explosive—Class A."

5.2.2 *Level B.* The M17, M18 and M24 containers shall be marked in accordance with dwg. 8858848. The M25 container shall be marked in accordance with dwg. 8858577.

5.2.3 *Level C.* The fiberboard carton shall be marked in accordance with dwg. 20–4–619.

5.3 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Packing required (see 5.1).

6.2 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists in accordance with the instructions of paragraphs 6.2.1, 6.2.2, 6.2.3, 6.2.4 and 6.2.5.

6.2.1 Inspection Equipment Lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment—00 Form 1242–8—the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design". Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraph 6.2.2 and 6.2.3. Action required of the contractor with respect to "commercial item" inspection equipment, is described in paragraph 6.2.4. The contractor will be required to prepare drawings for all equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND4.

6.2.2 *Detailed Drawings.* Detailed drawings are drawings which completely depict all of the information necessary for the fabrication of the item of inspection equipment. Contractors need provide no design when a detailed drawing is listed for an item of inspection equipment and the contractor elects to use the Government design. The contractor may, however, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND4.

6.2.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control

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drawings. The IEL code will indicate whether the drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND4.

6.2.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.2.5 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of Government or contractor designed gages, diagrammatic sketches, or other descriptions of the use of such equipment, shall be submitted to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND4 for approval.

6.3 **Inspection code numbers.** The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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